



Match the Event

Quick facilitation notes & answer key

Goal

Students read each script's hat and match it to its starting event — green flag vs a received message — and check that a broadcast MATCHES the receive hat. Matching the trigger to the script (and the message to the receiver) is reading code with events, building toward debugging mismatched messages later. This is Grade-2 C3.2 reading of code that involves concurrent events.

Ontario curriculum (Grade 2 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: cards labelled 'green flag' and 'jump'; students place the right card on top of each script.

Run it (about 20 min)

1. I DO: point to Bit's hat — 'green flag starts Bit.' Then Pixel's hat — 'a message called jump starts Pixel.'
2. WE DO: check the messages — Bit broadcasts 'jump', Pixel waits for 'jump'. Same word → they MATCH → Pixel runs.
3. YOU DO: students write each script's event, answer that the messages match, and find both landings.
4. Connect: 'A receive hat only fires if the broadcast message is exactly the one it waits for.'

Answer key (modelled by the run-gate)

Ex 1 — Bit's event is 'when green flag clicked'.

Ex 2 — Pixel's event is 'when I receive jump' (the message 'jump').

Ex 3 — Bit broadcasts 'jump' and Pixel waits for 'jump' → YES, they match → Pixel runs. Bit: 0 → forward 1 → 1. Pixel: 4 → back 1 → 3. Bit 1, Pixel 3.

Match note: if Bit had broadcast a different word, Pixel's hat would not fire and Pixel would stay home — the messages must be the same.

Watch for & differentiate

Common error: matching by sprite colour instead of by the hat — the EVENT is named in the orange hat, not by who is green or purple.

Common error: assuming any broadcast starts any receiver — only a matching message fires the receive hat.

Simplify: underline the message word in 'broadcast jump' and 'when I receive jump' and check they are the same.

Extend: change Pixel's hat to 'when I receive spin' and ask whether Bit's 'jump' broadcast still starts it (no).

Success indicator: the child names both events, confirms the messages match, and lands Bit on 1 and Pixel on 3.